

Literature Status: Fixed Point Spectrum for L_p Actions

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Status

This is a literature-already-answered packet, not an original proof from the run. Problems 14 and 15 in Nowak, arXiv:1302.6609, ask whether the L_p fixed point spectrum $\mathcal{F}(G)$ of a finitely generated property-(T) group is connected, and whether there is a critical value $p > 2$ with

$$\mathcal{F}(G) = (1, p).$$

Marrakchi–de la Salle, arXiv:2001.02490, answer these questions by proving that the complementary set of exponents admitting affine isometric L_p actions with unbounded orbits is an upper interval.

Original Problem

The original source is:

P. W. Nowak, *Group Actions on Banach Spaces*, arXiv:1302.6609.

On page 10, Nowak defines

$$\mathcal{F}(G) = \{p \in (1, \infty) : H^1(G, \pi) = 0 \text{ for every isometric representation } \pi \text{ on an } L_p\text{-space}\}.$$

The paper then asks:

- Problem 14 (C. Drutu): is $\mathcal{F}(G)$ connected?
- Problem 15: given G with property (T), is there a critical value $p > 2$ such that $\mathcal{F}(G) = (1, p)$?

Later Answer

The supporting source is:

A. Marrakchi and M. de la Salle, *Isometric actions on L_p -spaces: dependence on the value of p* , arXiv:2001.02490.

Their Theorem 1 shows monotonicity in p for affine isometric actions on L_p spaces. Corollary 2 then states that, for every topological group G , the set of $p \in (0, \infty)$ for which G admits a continuous affine isometric action on an L_p space with unbounded orbits has the form

$$(p_G, \infty) \quad \text{or} \quad [p_G, \infty)$$

for some $p_G \in \{0\} \cup [2, \infty]$. The same page says this answers the question raised by Chatterji–Drutu–Haglund, often referred to as Drutu’s conjecture.

For second countable locally compact groups with property (T), the supporting paper records the sharper endpoint information: $p_G > 2$, and there is an affine isometric L_{p_G} action without a fixed point. Finitely generated groups are second countable locally compact in the discrete topology, so this applies to Nowak’s stated setting.

Translation to Nowak’s Spectrum

For $1 < p < \infty$, L_p spaces are reflexive. In this range, an affine isometric action has a fixed point if and only if its orbits are bounded; this is recalled on page 2 of arXiv:2001.02490 and is also the fixed-point principle used in Nowak’s source. Therefore

$$p \in \mathcal{F}(G) \iff \text{there is no affine isometric } L_p \text{ action with unbounded orbits.}$$

Since the unbounded-orbit set is $[p_G, \infty)$ for property-(T) groups in Nowak’s class, its complement in $(1, \infty)$ is exactly

$$\mathcal{F}(G) = (1, p_G),$$

with $p_G > 2$. This proves connectedness and the critical-value form asked for in Problems 14 and 15.

Verification Notes

The answer is not contained in arXiv:1302.6609; that paper explicitly says the problems were far from understood. The later paper is independent literature and explicitly identifies Corollary 2 as an answer to Drutu’s conjecture. The only mathematical translation needed is the standard equivalence, for $1 < p < \infty$, between vanishing first cohomology for all isometric representations and fixed points for all affine isometric actions on L_p spaces.

Limitations

This packet does not compute the numerical value of p_G for any specific group. It also does not address the other questions in Nowak’s survey, including the $\tilde{A}_2$ question, uniformly bounded Hilbert representations, reduced ℓ_p cohomology, Yu’s question, or Nica’s mapping-class-group conjecture.

Human Review Recommendation

Verify the endpoint statement on page 2 of arXiv:2001.02490 and the fixed-point/bounded-orbit translation for affine isometric actions on L_p , $1 < p < \infty$. These two points are the complete bridge from the later paper to Nowak’s Problems 14 and 15.